Grant Poquiz | gpoquiz.me

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Experience

SOFTWARE DEVELOPER II - CLARITY SML RFID

November 2023 – Current

- Item-level inventorying with RFID technology for increased accuracy and speed.
- MVVM Dependency Injected Android App, Svelte Web App, C# API backend, Azure Devops Deployment, SQLServer Repository, Atlassian Suite (Jira)
- Optimized an endpoint from 20 minutes to 45 seconds within one week of starting.
- Personalized changes for large clients: H&M, Tesco, Nike, Academy, Altar'd State, and more.

Xamarin To Maui Refactor

- Worked with 4 devs to upgrade 250+ pages and controls to .NET9 and Maui.
- Created custom regex to accelerate process beyond expectations.

RFID Locationing Research and Development

- Generated, parsed, and analyzed RFID read data.
- Designed strength algorithm for translating hundreds of raw reads into a condensed distance value.

Order Cancellation Reason Codes

- Solo Project for H&M: Ask employees why they are canceling a customer's online order.
- Designed and implemented repository, API, and pages from scratch.

QuickPrint Promotions

- Solo Project for Altar'd State: Simplified and quickened workflow for tagging newly promoted items on a clearance rack.
- Modified existing repository, imports, and API calls. New pages from scratch.

SOFTWARE CONTRACTOR - AMBIT ENERGY SharpEcho

May 2022 – September 2023

- DevOps and Web development work for the Ambit Energy team at VistraCorp.
- Consultant Marketing Portals, Client Management, Client Enrollment, etc.
- C# Fabric/ASP.NET Micro-Services Backend, SQL Server Database Management, Vue.js and Angular front end.

Universal Exception Queue

- Transformed an existing Ambit case system into a high-performance Universal Exception Queue.
- Implemented hashing for unique case identification, reduced duplicate cases, and optimized indexing for faster query responses.

Ambit Payment Extensions

- Interteam project for seamlessly extending a payment processing system for agents.

SOFTWARE DEVELOPER II Paycom

June 2021 - May 2022

- Full stack development working with Paycom's largest module: Payroll.
- PHP backend, Javascript frontend, and mySQL/FoxPro database management for high profile clients such as Amazon, The New England Patriots, and Planet Fitness
- Generation and management of W2's, 1099's, and other tax forms with C# .NET

Automated SUTA Rate Updates

- Leveraged PHP, MySQL, RabbitMQ, and Orchestrator Architecture to streamline notifying clients about updated tax rates.

Payroll Late Submission Reminders

- Made custom PHP pages and MySQL databases to create and manage reminders of late payroll submissions.

Education

UNIVERSITY OF TEXAS AT DALLAS

Master of Science in Computer Science and Interactive Computing
Bachelor of Science in Computer Science

Projects

COLLAPSI AI

C++

- Created a Multi-Threaded C++ solver and analyzer for a card game: Collapsi by Mark Ball.
- Calculates 52,000,000 end states over 10,000 games. 3400 states/ms, 1.5ms/game
- Modified at Game Designer's request to calculate end score given different solve depths.
- Modified performance ranged from .01 ms/game with depth 1 to 1 ms/game with maximum depth

DIGITAL COLLAPSI

C++, GDScript, GDExtension, Godot Game Engine

- Developed a digital version of Collapsi using Godot.
- Custom Spriting, Foley/Sound Effects, and Code
- Ported the C++ solver as a GDExtension to utilize it as an AI opponent.
- Play online or download for free here: gpoquiz.itch.io/collapsi

DUTCHMAN'S DELVE

C#, Unity

- Designed, developed, and polished a physics based platformer with a partner for Brackey's One Week Game Jam.
- Download for free: anemonestudio.itch.io/dutchmans-delve

UNREAL CHASM GENERATION

C++, Unreal, Nanite, Megascans, SFML

- Generated Voronoi Diagrams to create chasmous landscapes.
- Experimented with SDL2, SFML, and Cairo.
- Created custom height-maps that use 8 bit RGB instead of 16 bit gray-scale.
- Converted the Voronoi Diagrams into splines with high-definition megascans and intelligent modifications to create procedural geometry.

MULTIDIMENSIONAL CELLULAR AUTOMATA

C++, Unreal

- Simulated cellular automata (games of life) in generalized dimensions with Unreal Engine.

CUDA CELLULAR AUTOMATA

C++, CUDA, SDL

- Optimized the generation of N'th dimensional cellular automata with CUDA.

VIRTUAL REALITY QUIZ

C#, Unity Game Engine, Photon Multiplayer, GoogleVR

- Created online multiple choice and geo-guesser style multiplayer quizzes.
- Implemented a Peer-to-Peer Host-Client architecture using Remote Procedure Calls.

VR GRAVITY MANIPULATION

C#, Unity Game Engine, Virtual Reality

- Devised a system of locomotion in virtual reality where players would point to manipulate their direction of gravity.

VR DOUBLE VISION

C++, Unreal Blueprints, OpenXR

- Manipulated materials in Unreal Engine 5 to make specific objects only visible to one eye, mimicking an anaglyph image.

SIGNAL ANALYSIS

C++, PortAudio, Audio Analysis

- Analyzed microphone input with the low-level PortAudio library in order to isolate tone for mimicking rock-band style gameplay.

Skills

Simulation | Research | Compilers | Language Learning | Verilog | C | RFID | Signal Processing